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A selection of the topics from my undergraduate coursework at PES University that I rely on most in research. It is not an exhaustive syllabus. I have grouped it by subject and put the mathematics first, since everything else rests on it.

Linear algebra

- **Systems and factorizations.** The geometry of linear systems, Gaussian elimination, LU and Cholesky factorization, and inverses by Gauss–Jordan.
- **Vector spaces and linear maps.** Subspaces, linear independence, basis and dimension, the four fundamental subspaces, the rank–nullity theorem, linear transformations and their algebra, invertible maps and isomorphisms.
- **Orthogonality.** Orthogonal bases, projections and least squares, and the Gram–Schmidt process.
- **Spectral theory.** Eigenvalues and eigenvectors, the Cayley–Hamilton theorem, the spectral theorem and diagonalization of symmetric matrices, quadratic forms, and positive definite and semidefinite matrices.
- **Singular value decomposition.** The SVD and the pseudoinverse, with applications.

Probability and statistics

- **Probability.** Random variables, expectation and functions of random variables; the Bernoulli, binomial, Poisson and normal distributions and how they are derived; generating random variates; Chebyshev's inequality.
- **Limit theorems and sampling.** Sampling distributions and the central limit theorem, with applications.
- **Estimation.** Principles of point estimation: mean squared error and maximum likelihood estimates for Bernoulli, binomial, Poisson and normal models.
- **Inference.** Confidence intervals for means, proportions and paired data, including the Student's t distribution; hypothesis tests for means and proportions; Type I and Type II errors and the power of a test; chi-squared and other distribution-free tests.
- **Regression.** Simple and multiple linear regression, least-squares lines, uncertainty in the coefficients, and checking assumptions.
- **Data analysis.** Sampling methods and errors, controlled and observational studies, and data visualization and interpretation.

Calculus and differential equations

- **Multivariable calculus.** Partial and total derivatives, the chain rule, Euler's theorem for homogeneous functions, Taylor and Maclaurin series in two variables, maxima and minima, and Lagrange multipliers.
- **Multiple integrals.** Jacobians, change of order and change of variables for double and triple integrals, including spherical and cylindrical coordinates.
- **Ordinary differential equations.** First-order equations (Bernoulli, exact, orthogonal trajectories, nonlinear), higher-order linear equations with constant and variable coefficients, variation of parameters, and Legendre's equation.
- **Partial differential equations.** Formation of PDEs, Lagrange's linear equation, and separation of variables.
- **Special functions.** The beta and gamma functions, and Bessel functions: series solutions, recurrence relations, generating functions and orthogonality.

Transforms and harmonic analysis

- **Laplace transforms.** Transforms of elementary, periodic and unit-step functions, inverse transforms, the convolution theorem, and solving linear differential equations.
- **Fourier series.** Dirichlet's conditions, Euler's formulae, even, odd and half-range series, complex Fourier series, and Parseval's identity.
- **Fourier transforms.** Finite and infinite transforms, sine and cosine transforms, their properties and inversion. The course also included a MATLAB unit.

Graph theory and network science

- **Structure.** Walks, paths and circuits, Euler and Hamiltonian graphs, trees and spanning trees, fundamental circuits, cut sets, connectivity and separability, isomorphism, and planar graphs.
- **Flows.** Network flows and the max-flow min-cut theorem.
- **Coloring and matching.** Chromatic number, partitioning and polynomial, matchings and coverings, the Four Color problem, and register allocation by graph coloring.
- **Algorithms.** Minimum spanning trees (Borůvka), shortest paths (Bellman–Ford, Floyd–Warshall, Johnson), articulation points, strongly connected components, the Chinese postman problem, and timetabling.
- **Network science.** Centrality measures (degree, betweenness, closeness, eigenvector, Katz, PageRank), clustering and transitivity, components and cohesive substructures, and community detection (clique percolation, Girvan–Newman, Louvain).
- **Network models and dynamics.** Random graph, small-world (Watts–Strogatz) and scale-free (Barabási–Albert) models, myopic search, diffusion and epidemic models, cascades, and percolation and robustness.
- **Graph databases.** Neo4j and its query language, hands-on.

Theory of computation and logic

- **Finite automata and regular languages.** Deterministic and nondeterministic finite automata and their equivalence, state minimization, regular expressions and grammars, closure properties, and identifying non-regular languages.
- **Context-free languages.** Pushdown automata and context-free grammars, ambiguity and normal forms, a membership algorithm, closure properties, and the pumping lemma.
- **Computability.** Turing machines and Turing's thesis, the Chomsky hierarchy, recursive and recursively enumerable languages, undecidable problems, and the idea of reduction.
- **Logic.** Propositional and first-order logic: syntax and semantics, and theorem proving by resolution and conjunctive normal form.

Machine learning

- **Foundations.** Hypothesis spaces, version space and inductive bias; performance metrics (precision, recall, ROC and AUC) and the bias–variance decomposition.
- **Supervised learning.** Decision trees (ID3) and overfitting, linear and logistic regression with gradient descent, k-nearest neighbors, random forests and gradient boosting, and support vector machines via margin maximization and the Lagrangian dual (hard margin).
- **Neural networks.** Perceptrons, multilayer networks and backpropagation, activation functions, and optimizers (SGD, momentum, Adagrad, Adam).
- **Stochastic models and unsupervised learning.** Bayes' theorem, maximum likelihood, naive Bayes and the Bayes optimal classifier, expectation–maximization and Gaussian mixtures, hidden Markov models, and clustering (hierarchical, and k-means as a special case of EM).
- **Deep learning.** Convolutional and recurrent networks, vanishing and exploding gradients, LSTMs and GRUs, GANs, and the Transformer with attention.

Robotics and autonomous systems

- **Hardware and software.** Sensors and actuators, and ROS and ROS 2: their architecture, communication, and packages for hardware control.
- **Locomotion.** Ground robots, with wheel types and arrangements and forward and inverse kinematics; aerial robots, with the dynamics of thrusters and propellers.
- **Sensing.** Gyroscopes, accelerometers, IMUs, GPS, range sensors, cameras and LiDAR.
- **Robot vision.** Image processing and filtering, object detection and tracking, stereo vision and 3D perception, and scene segmentation.
- **Localization and SLAM.** Belief and map representations, probabilistic map-based localization (Markov and Kalman filter localization), and SLAM: EKF SLAM, particle filter SLAM, GraphSLAM and MonoSLAM.
- **Navigation.** Path planning by graph search and potential fields, obstacle avoidance (Bug algorithm, vector field histogram, bubble band, curvature velocity and dynamic window approaches), and navigation with reinforcement learning and imitation learning.

Databases

- **Data modeling.** The entity–relationship model and reducing it to a relational schema, keys and constraints.
- **Relational algebra and SQL.** Relational algebra operations, SQL queries with nested subqueries and joins, views, triggers, functions and procedures, and an introduction to query processing and optimization.
- **Design theory.** Functional dependencies, inference rules, closure and minimal cover, normal forms (1NF, 2NF, 3NF and Boyce–Codd), and properties of relational decompositions.
- **Transactions.** Database transactions, concurrency control, locking and recovery, and database security.
- **NoSQL databases.** Document (MongoDB), key-value (DynamoDB), graph (Neo4j) and wide-column (HBase) stores.

Compilers

- **Lexical analysis.** Specifying and recognizing tokens with regular expressions and finite automata, input buffering, and lexer generators.
- **Parsing.** Context-free grammars, ambiguity, left recursion and left factoring; top-down parsing (recursive descent, LL(1)) and bottom-up parsing (shift–reduce, LR(0), SLR, CLR, LALR); and YACC.
- **Semantics and intermediate code.** Syntax-directed definitions and translation, syntax trees and DAGs, three-address code, SSA form, and control-flow graphs.
- **Optimization and code generation.** Data-flow analysis (live-variable analysis), local, global and loop optimizations, run-time environments and stack allocation, and the design of a code generator.